

# A Modified Hamming Distance Measure for Quick Detection of Dissimilar Binary Images

Adnan A. Y. Mustafa  
Department of Mechanical Engineering  
Kuwait University  
Kuwait  
adnan.mustafa@ku.edu.kw

**Abstract**—Given two dissimilar binary images we are interested in detecting their dissimilarity quickly. In our previous work, we developed a Probabilistic Model for Matching (PMM) that can be used to detect distinct-dissimilar binary images quickly with high confidence; e.g., by checking as few as five points with 93% confidence. In this paper we develop a similarity measure to be used with PMM. The similarity measure is a modification of the Hamming distance that measures the concurrence between two images. We show using real images, that by using this similarity measure, PMM can be extended to quasi-similar (non-distinct-dissimilar) images.

**Keywords**—similarity distance, similarity measure, image matching and binary images.

## I. INTRODUCTION

Binary images appear in many aspects of image analysis. The image might have been acquired as a binary image, or converted from a color or grayscale image, but usually it is a product of a segmentation process. In many cases we are interested in matching the binary image or part of it. While many techniques exist for matching, they all suffer from a major handicap; as image size increases, processing time also increases. With 20 mega-pixel images common today and easily producible by inexpensive digital cameras or mobile phones, and with 50 mega-pixel cameras recently introduced in the markets [1] the need to quickly match huge images is becoming a major challenge.

In [2] we presented a framework for quick rejection of dissimilar binary images and continued our development of a Probabilistic Matching Model (PMM) in [3] that is magnitudes faster than other existing area-based methods in use today [4]. However, an adequate similarity measure is in need to complete the development of the model. In this paper, we present such a similarity measure, called the binary similarity measure ( $\gamma$ ), that can be integrated with the model and used to measure how fast the model will detect dissimilarity between different images.

This paper is organized as follows: section II presents an overview of common similarity measures and distances cited in the literature. Section III discusses image closeness and presents a brief overview of PMM. Section IV presents the development of  $\gamma$  and its similarities and differences with the Hamming distance and other measures. Section V discusses the application of  $\gamma$  with examples and the use of the similarity measure with PMM. Finally section VI presents our conclusions and future work.

## II. SIMILARITY MEASURES AND DISTANCES

Many similarity measures for comparing binary attributes of objects have been introduced over the last century. Here, we briefly review some of these measures and distances that are applicable to binary attributes and are relevant to our work in matching binary images. In our discussion below,  $p$  and  $q$  are object vectors that consist of  $n$  binary attributes.

One of the earliest similarity measure introduced was by Jaccard in 1901 [5]. If  $M_{ij}$  is the number of attributes where  $p = i$  and  $q = j$ , with  $i, j \in \{0,1\}$ , then the Jaccard coefficient ( $J$ ) is defined as,

$$J = \frac{M_{1,1}}{M_{0,1} + M_{1,0} + M_{1,1}} \quad (1)$$

which represents the ratio of the number of matches present to the total number of attributes present (i.e. with the absence of attributes term,  $M_{0,0}$ , removed). The Jaccard distance ( $d_J$ ), also neglects the  $M_{0,0}$  term, and is defined by  $d_J = 1 - J$ . A value of  $J = 1$  (corresponding to  $d_J = 0$ ) indicates complete similarity between two objects, while a value of  $J = 0$  indicates no similarity. Because Jaccard neglects the  $M_{0,0}$  term, the Jaccard coefficient and distance are used to handle objects consisting of asymmetric binary attributes.

A very common similarity measure used, that does not neglect the  $M_{0,0}$  term, is the Simple Matching Coefficient (SMC) [6] defined as,

$$SMC = \frac{M_{0,0} + M_{1,1}}{M_{0,0} + M_{0,1} + M_{1,0} + M_{1,1}} \quad (2)$$

The Simple Matching Distance (SMD) is then defined as  $SMD = 1 - SMC$ . A value of  $SMC = 1$  indicates complete similarity between two objects, while a value of  $SMC = 0$  indicates no similarity.

Another popular similarity measure, particularly efficient for sparse vectors, is the Cosine similarity [7] defined as,

$$Cos = \cos(\theta) = \frac{p \bullet q}{\|p\| \bullet \|q\|} \quad (3)$$

The range of the cosine similarity is  $[-1,1]$ , where a value of  $-1$  means exact inverse similarity and a value of  $1$  means exact similarity. A value of  $0$  indicates independence. Pearson's correlation coefficient [8] is a statistical measure of the linear relationship between the attributes of the vectors.

The range of the correlation is  $[-1, 1]$ , where a value of  $-1$  implies perfect inverse linear relationship and a value of  $1$  means perfect linear relationship. A value of  $0$  indicates linear independence (but not necessarily non-linear independence). Using correlation for image matching is a popular approach.

The *Hamming distance*, originally developed for non-binary attributes [9] reduces to the following for binary data,

$$d_H = \sum_{i=0}^n (p_i \oplus q_i) \quad (4)$$

where the  $\oplus$  symbol represents the logical *exclusive or* operation (usually denoted by *XOR*). Note that for binary data, if the *Hamming distance* is normalized with respect to the number of attributes, then the *SMD* similarity distance is produced.

Joint entropy [10] between two binary vectors can also be used as a similarity measure. Entropy is a measure of the information embedded in a vector. The joint entropy of two vectors measures the amount of information between two vectors. Let  $E(p, q)$  denote the joint entropy between  $p$  and  $q$  given by,

$$E(p, q) = - \sum_{i=0}^1 \sum_{j=0}^1 h_{ij}(p, q) \log_2(h_{ij}(p, q)) \quad (5)$$

where  $h_{ij}(p, q)$  for  $i, j = 0, 1$  denotes the joint normalized histogram distribution of the two vectors  $p$  and  $q$ , with  $0 \leq E(p, q) \leq 2$ . Hence, a normalized joint entropy is defined as,

$$NJE(p, q) = 2 - \frac{E(p, q)}{\max(E(p, p), E(q, q))} \quad (6)$$

Hence,  $0 \leq NJE(p, q) \leq 1$ . A value of  $NJE(p, q) = 1$  implies that the vectors are the same. Unfortunately, computing entropy is intensive because of the multiple calls to the  $\log$  function.

Other well-known similarity non-distances can be used to measure similarity with minor adjustments, e.g. *Minkowski distance* [11], *Mahalanobis Distance* [12], *Hausdorff distance* [13]. Additional binary similarity measures can be found in [14].

### III. A PROBABILISTIC MODEL FOR MATCHING

The *PMM* theorem and its corollaries were introduced in [17]. Here we review *PPM* by first discussing how it categorizes image closeness and then summarizing the *PPM* theorem.

#### A. Image Closeness

The closeness between two images is based on a pixel-to-pixel comparison. This closeness is categorized as either similar or dissimilar:

- *Similar (S)*: The two images are considered to be the same and are of two types:
  - 1) *Exact (E)*: The two images are exactly the same. This can only be true if the two images have the same intensity values at all corresponding pixels.

- 2) *Inverse (I)*: The two images are the inverse of each other. This can only be true if the two images have the complement intensity values at all corresponding pixels.

- *Dissimilar (R)*: The two images are different and this can only be true if they are not similar; i.e. neither exact nor inverse. They are of two types:

- 1) *Distinct-Dissimilar (D)*: The two images are completely different (as explained below).
- 2) *Quasi-Dissimilar (or Quasi-Similar) (Q)*: The two images have some resemblance between them.

#### B. The PMM Theorem

The *PMM* theorem and its corollaries can be summarized as follows:

- By mapping random corresponding points between any two distinct-dissimilar binary images, the probability of the images being dissimilar on the  $p^{\text{th}}$  mapping,  $P(D, p)$ , is given by,

$$P(D, p) = 1 - \frac{1}{2^{p-1}} \quad p \geq 1 \quad (7)$$

- $P$  can be interpreted as the confidence level or detection confidence (*DC*) that an image pair is indeed dissimilar at a given mapping  $p$ .
- Given any two distinct-dissimilar binary images, the number of random mappings required to ascertain that the two images are dissimilar with probability  $P$  is,

$$p(D, P) = 1 - \log_2(1 - P) \quad 0 \leq P \leq 1 \quad (8)$$

- As the similarity for any quasi-similar image pair decreases and approaches zero, the image pair becomes closer to being distinct-dissimilar.
- For any probability  $P$ , the mean number of random mappings,  $p$ , required to ascertain that any two binary images are quasi-similar, cannot be less than the mean number of mappings required to ascertain that any two binary images are distinct-dissimilar.
- For any probability  $P$ , the mean number of random mappings,  $p$ , required to ascertain that any two binary images are quasi-similar, increases as the images become more similar.

By applying *PMM* fast detection of distinct-dissimilar binary images is possible and image size is not an issue. The *Dissimilar Detection via Mapping* method (*DDM*) is a method based on *PMM* that can detect dissimilar binary images quickly. For example, it has been found to be over 40,000 times faster in detecting dissimilar images than the fastest method in use today, when processing 10 mega-pixel images. However, the equations stated above are only applicable to distinct-dissimilar images, and not for quasi-similar image pairs. Nevertheless, these equations can still be used as a reference for quasi-similar images as stated by the *PMM* theorem; the required number of mappings will increase depending on how far off the image pairs are from being distinct-dissimilar. Hence, if a similarity measure that

adequately measures this offset is available, the required number of mappings can then be approximated. The development of such an adequate similarity measure is the focus of this paper.

#### IV. THE $\gamma$ BINARY SIMILARITY MEASURE

For our specific purpose, we need a similarity measure that,

- takes into account that images are similar whether they are exact or inverse,
- agrees with the other image categories stated above,
- is directly related to the percentage of either intensity value,
- is easily quantified and mathematically measurable, and
- is applicable to the PMM model.

In our quest to find such a binary measure, we found that the existing binary measures and distances were not adequate for our work. Hence, a new similarity measure was developed that later was recognized to be nothing more than a modification to the Hamming distance, but with different properties as we show below.

##### 1) Definition of the Binary Similarity Measure

The binary similarity measure ( $\gamma$ ) measures the amount of similarity and concurrence between two images. Formally, given two images  $\mathbf{u}$  and  $\mathbf{v}$ ,  $\gamma$  is defined as,

$$\gamma(\mathbf{u}, \mathbf{v}) = \left| 1 - \frac{2}{MN} \sum_{r=1}^M \sum_{c=1}^N f_{\gamma}(u_{r,c}, v_{r,c}) \right| \quad \mathbf{u} \in \mathbf{u}, \mathbf{v} \in \mathbf{v} \quad (9)$$

where

$$f_{\gamma}(u, v) = u + v - 2uv \quad (10)$$

and  $M \times N$  is the image size. The double summation term in the above equation is equivalent to the value of the probability mass function of the image ( $p_o$ ) produced by applying the  $\oplus$  operation to the two images, i.e.,

$$\frac{1}{MN} \sum_{r=1}^M \sum_{c=1}^N f_{\gamma}(u_{r,c}, v_{r,c}) = p_o((\mathbf{u} \oplus \mathbf{v}) = 1) \quad (11)$$

i.e., (9) can be rewritten as,

$$\gamma(\mathbf{u}, \mathbf{v}) = |1 - 2p_o((\mathbf{u} \oplus \mathbf{v}) = z)| \quad z \in \{0, 1\} \quad (12)$$

Here  $z$  can be either binary value; both values produce the same result, since  $\gamma$  is symmetric and,

$$p_o(\mathbf{x} = 1) = 1 - p_o(\mathbf{x} = 0) \quad \mathbf{x} \in \{0, 1\} \quad (13)$$

For any  $\mathbf{x}$ ,  $p_o(\mathbf{x})$  is measured by the normalized histogram of  $\mathbf{x}$ . Applying an XOR operation to two images produces a segmented image that discriminates between where the pixel intensities of the two images agree or differ. As a result,  $h(\mathbf{x} = z)$  represents the percentage of pixels with intensity  $z$  and  $\gamma$  is a measure that represents the percentage of image pixels that are the same (or the complement).

Fig. 1 shows a graph of the  $f(x) = |1 - 2x|$  function implied in the definition of  $\gamma$ . If the value of  $x$  is restricted to be in the range  $[0, 1]$ , then the function is symmetric about its absolute

minimum value at  $x = 0.5$ . The function is even, nonnegative and attains a maximum value of unity at its border values of  $x = 0, 1$ . Because the function is symmetric, any value of  $x$  a distance of  $\pm d$  from its minimum has the same value,

$$f(\frac{1}{2} - d) = f(\frac{1}{2} + d) \quad d \in [0, \frac{1}{2}] \quad (14)$$

The  $f(x)$  function is fitting for our purposes, as it will produce for any image and its complement the same value.

##### 2) $\gamma$ and the Hamming distance

If the Hamming distance ( $d_H$ ) is applied to binary vectors  $\mathbf{u}$  and  $\mathbf{v}$ , and normalized, we obtain,

$$\frac{1}{n} d_H(\mathbf{u}, \mathbf{v}) = \frac{1}{n} \sum_{i=0}^n (u_i \oplus v_i) \quad (15)$$

This can be equivalently rewritten as,

$$\frac{1}{n} d_H(\mathbf{u}, \mathbf{v}) = p((\mathbf{u} \oplus \mathbf{v})) = 1 \quad (16)$$

Hence, the binary similarity measure ( $\gamma$ ) can be defined in terms of the Hamming distance as,

$$\gamma(\mathbf{u}, \mathbf{v}) = \left| 1 - \frac{2}{n} d_H(\mathbf{u}, \mathbf{v}) \right| \quad (17)$$

However, unlike the Hamming distance,  $\gamma$  is not a metric; while it satisfies the positive-definite criteria ( $\gamma(u, v) \geq 0$ ) and symmetry ( $\gamma(u, v) = \gamma(v, u)$ ) criteria, it does not (always) satisfy the triangle inequality criteria,  $\gamma(u, v) + \gamma(u, w) \geq \gamma(v, w)$ . Hence, it is possible to have  $\gamma(u, v) + \gamma(u, w) < \gamma(v, w)$ . As an example, for the three images shown in Fig. 2,  $\gamma(I_1, I_2) = 0.255$ ,  $\gamma(I_1, I_3) = 0.022$ , and  $\gamma(I_2, I_3) = 0.122$ , which clearly violates the triangle inequality criteria.

##### 3) $\gamma$ and Joint Entropy

The normalized joint entropy of two binary images,  $\mathbf{x}_1$  and  $\mathbf{x}_2$  has some common characteristics with  $\gamma$ . An important one is that an image and its complement will produce the same result for both measures,

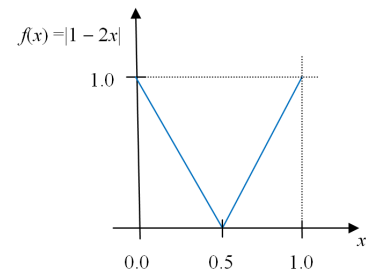


Figure 1. A graph of  $f(x) = |1 - 2x|$ .



Figure 2. Three sample images; from left to right,  $I_1$ ,  $I_2$  and  $I_3$ .

$$NJE(\mathbf{x}, \mathbf{x}) = NJE(\mathbf{x}, \neg \mathbf{x}), \quad \gamma(\mathbf{x}, \mathbf{x}) = \gamma(\mathbf{x}, \neg \mathbf{x}) \quad (18)$$

It is one of the few similarity measures that produce similar results for such cases.

4) *Categorization of Image Closeness based on  $\gamma$*   
 $\gamma$  has the range  $0 \leq \gamma \leq 1$ . Values of  $\gamma$  correspond to,

- $\gamma = 0$  for distinct-dissimilar images
- $0 < \gamma < 1$  for quasi-similar images
- $\gamma = 1$  for *similar* images

We further categorize similarity between images into the following 7 categories as shown in Fig. 3:

1. Distinct-dissimilar images (no similarity):  $\gamma = 0$ .
2. Minimal similarity:  $0 < \gamma \leq 0.2$ .
3. Low similarity:  $0.2 < \gamma \leq 0.4$ .
4. Medium similarity:  $0.4 < \gamma \leq 0.6$ .
5. Good similarity:  $0.6 < \gamma \leq 0.8$ .
6. High similarity:  $0.8 < \gamma < 1$ .
7. Similar images (perfect similarity):  $\gamma = 1$ .

In practice, image pairs with  $\gamma < 0.01$  are assumed to be  $\gamma \approx 0$ , and are considered to be distinct-dissimilar image pairs.

## V. DISCUSSION

In this section we give examples of  $\gamma$  with real images. We also compare its performance to some of the other similarity measures discussed. Finally we show how it can be used with PMM.

### A. Sample Images

Fig. 4 shows a set of image pairs of similar shapes and their associated  $\gamma$  value. From a first glance, it can be seen that the images are of high similarity and hence  $\gamma$  values should be high. The result of XOR-ing each image pair is shown next to the image; mismatched locations for each image pair are shown in white. The  $\gamma$  value for each pair is given in the caption of the figure. The amount of dominance of either binary value in the XOR image is what controls the value of  $\gamma$ . For example, the lowest  $\gamma$  values are for the square and thin rectangle with a value of  $\gamma = 0.651$  (3<sup>rd</sup> pair of the left column) and for the thin horizontal rectangle and a non-thin rectangle with a value of  $\gamma = 0.648$  (2<sup>nd</sup> pair of the right column). The largest  $\gamma$  values are for the square and non-thin rectangle with a value of  $\gamma = 0.851$  (2<sup>nd</sup> pair of the left column) and for the short rectangle with a long rectangle with a value of  $\gamma = 0.900$  (4<sup>th</sup> pair of the right column).

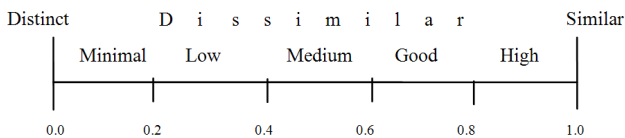


Figure 3. Image similarity classification based on  $\gamma$  value.

### B. Real Image Examples

Fig. 5 shows several image pairs and the result of applying the exclusive-or operation along with their  $\gamma$  value. The first image pair is for two real images that are very different, where nearly half of the corresponding pixels have different intensity values. This results in a value of  $\gamma = 0.002$  for this pair. Hence, this image pair is categorized as distinct-dissimilar. The 2<sup>nd</sup> image pair has 61.1% of the corresponding pixels with the same intensity value. Hence,  $\gamma = 0.222$  for this pair. Hence, this image pair is categorized as quasi dissimilar with low similarity. The 3<sup>rd</sup> image pair has 26.7% of the corresponding pixels with the same intensity value (which implies that 73.3% of the corresponding pixels have the inverse intensity value), and  $\gamma = 0.467$ . Hence, this image pair is categorized as quasi dissimilar with medium similarity. The 4<sup>th</sup> image pair has 83.5% of the corresponding pixels with the inverse intensity value, and  $\gamma = 0.669$  for this pair. Hence, this image pair can be categorized as quasi-similar with good similarity. The last image pair are identical, and  $\gamma = 1.000$  for this pair. Hence, this image pair is categorized as similar (exact).

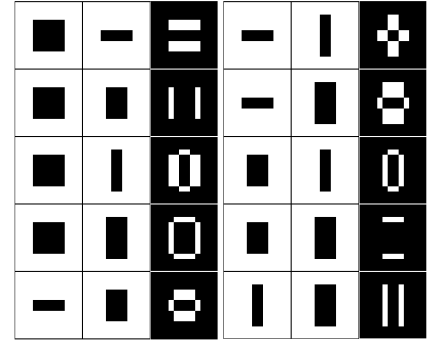


Figure 4. Image pairs and their XOR product. Left column: from top to bottom:  $\gamma = 0.702$ ,  $\gamma = 0.851$ ,  $\gamma = 0.651$ ,  $\gamma = 0.751$  and  $\gamma = 0.749$ . Right column:  $\gamma = 0.744$ ,  $\gamma = 0.648$ ,  $\gamma = 0.8$ ,  $\gamma = 0.9$  and  $\gamma = 0.802$ .



Figure 5. Sample image pairs and their XOR product.

### C. Comparison with Other Similarity Measures

Fig. 6 shows four images labeled  $I_1$ ,  $I_2$ ,  $I_3$  and  $\neg I_3$  ( $\neg$  denotes binary complement). Table I displays the results of matching seven image pairs and measuring their similarity using the following methods: *SMD*, *corr*, *NJE* and  $\gamma$ ,

- When an image was matched to itself: ( $I_3, I_3$ ), all four methods correctly produced the maximum similarity measure; a value of 1 for  $\gamma$ , *corr* and *JE* and a value of 0 for *SMD*.
- When an image was matched to its binary compliment: ( $I_3, \neg I_3$ ), three methods produced the minimum similarity measure; a value of -1 for *corr*, while  $\gamma$  and *SMD* produced a value of 1. *NJE* produced a maximum similarity measure of 1, which indicates that it does not recognize an image and its compliment as being dissimilar.
- Values of *NJE* and  $\gamma$  differ slightly in agreement with the comments above.
- It is noteworthy to mention that taking the absolute value of the similarity values produced by *corr* and comparing them to *NJE*, we see that they are very close to each other. Furthermore, values produced by *SMD* are very different from those produced by *NJE* as evident by Fig. 7 which is a plot of the similarity values of the previous table.

### D. Application of $\gamma$ with PMM

The primary motive behind the development of  $\gamma$  is to employ it with *PMM*. In this section we show how it can be used. Fig. 8 shows real images used in our work. The set consists of the following labeled images: 1) *urban*, 2) *mill*, 3) *for1*, 4) *micro*, 5) *for2*, 6) *trees*, 7) *phone*, 8) *coins*, 9) *cars*, 10) *video*, 11) *daisy* and 12) *park*. Table II shows  $\gamma$  values for all image pairs in the set. Some comments about  $\gamma$ ,

- The image set has a similarity range of  $0.002 \leq \gamma \leq 0.669$ . Hence, the image similarity of the set ranges from no similarity (distinct-dissimilar) to good similarity.
- The image set has a  $\gamma$  mean and standard deviation of 0.231 and 0.161, respectively.
- Two image pairs were found to be distinct-dissimilar ( $\gamma < 0.01$ ): the *urban-parking* image pair with  $\gamma = 0.002$  and the *mill-phone* image pair with  $\gamma = 0.005$ .
- The *forest1* and *coins* images are the most similar images in the set with  $\gamma = 0.667$ .



Figure 6. Four sample images:  $I_1$ ,  $I_2$ ,  $I_3$  and  $\neg I_3$ .

TABLE I. SIMILARITY MEASUREMENTS FOR IMAGES OF FIG. 6.

| Method      | Image Pair |            |                 |            |                 |                 |            |
|-------------|------------|------------|-----------------|------------|-----------------|-----------------|------------|
|             | 1          | 2          | 3               | 4          | 5               | 6               | 7          |
| <i>SMD</i>  | $I_1, I_2$ | $I_1, I_3$ | $I_1, \neg I_3$ | $I_2, I_3$ | $I_2, \neg I_3$ | $I_3, \neg I_3$ | $I_3, I_3$ |
| <i>corr</i> | 0.479      | 0.556      | 0.444           | 0.413      | 0.587           | 1.000           | 0.000      |
| <i>NJE</i>  | 0.044      | -0.124     | 0.124           | 0.058      | -0.058          | -1.000          | 1.000      |
| $\gamma$    | 0.059      | 0.156      | 0.156           | 0.095      | 0.095           | 1.000           | 1.000      |
|             | 0.043      | 0.111      | 0.111           | 0.173      | 0.173           | 1.000           | 1.000      |

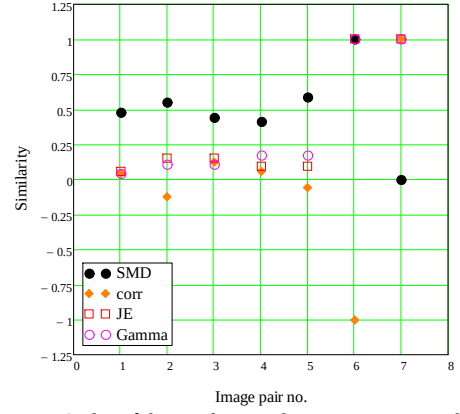


Figure 7. A plot of the similarity values appearing in Table 2.



Figure 8. Test images. Left to right and from top to bottom: First row: *urban*, *mill*, *for1* and *micro*. Second row: *for2*, *trees*, *phone* and *coins*. Last row: *cars*, *video*, *daisy* and *park*.

While one might expect that the *forest1* and *forest2* image pair should be the most similar with the highest  $\gamma$  (since they are image scenes of a forest), but this is not true. This would be true if  $\gamma$  compared the spatial relation of the pixels appearing in the scene, which  $\gamma$  does not do;  $\gamma$  only measures the concurrence between corresponding pixel values and does not “look” for spatial relations.

Fig. 9 shows a plot of  $\gamma$  vs. joint entropy for the image pairs of the images. We see that a strong negative correlation exists between  $\gamma$  and the joint entropy with a value of -0.767.

Let *MRN* denote the *Mapping Rejection Number* which is the number of mappings required to reject a pair of images as being dissimilar and *DC* denote the *Detection Confidence* which represents the expected (or requested) percentage of detected dissimilar images. Each pair of images in the set was matched against each other using *DDM* and the *MRN* for

different  $DC$  values with respect to  $\gamma$  was analyzed. A plot of  $MRN$  vs.  $\gamma$  for different  $DC$  values is shown in Fig. 10. From this figure, we see that when the values of  $\gamma$  are close to zero (i.e. distinct-dissimilar image pairs) the results are in agreement with (7). However, as  $\gamma$  increases, i.e. for quasi-similar image pairs, values for  $MRN$  for all  $DC$  values increases (non-linearly). This clearly shows the dependence of  $MRN$  on  $\gamma$  for quasi-similar image pairs. As an example, the *forest1* and *coins* images with the highest  $\gamma$  value with  $\gamma = 0.667$ , have the highest  $MRN$  value at  $DC = 0.999$ , with  $MRN = 40$ .

## VI. CONCLUSION

In this paper, we presented a binary similarity measure ( $\gamma$ ) that is a modification to the Hamming distance.  $\gamma$  measures the amount of similarity and concurrence between two images on a pixel-to-pixel basis and has values in the range  $[0,1]$ ;  $\gamma = 0$  for distinct-dissimilar image pairs,  $0 < \gamma < 1$  for quasi-similar image pairs and  $\gamma = 1$  for similar image pairs.  $\gamma$  is suitable for use with  $PMM$ ; tests show that the model equations are dependent on  $\gamma$ . Our future work will concentrate on extending  $PMM$  to quasi-similar image pairs and to develop a general equation for the prediction of the number of mappings required to detect quasi-similar image pairs as a function of  $\gamma$ .

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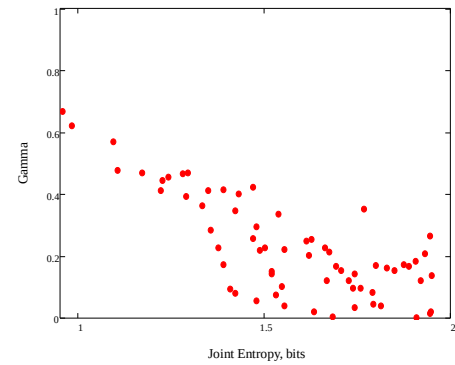


Fig. 9. A plot of  $\gamma$  vs. joint entropy for the image set.

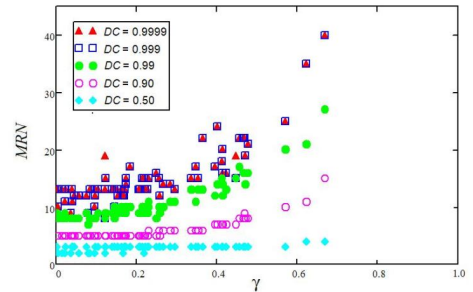


Fig. 10.  $MRN$  vs.  $\gamma$  for different  $DC$  values.

TABLE II.  $\gamma$  VALUES FOR THE REAL IMAGES SET

| Images | urban | mill  | for1  | micro | for2  | trees | phone | coin  | cars  | video | daisy | park  |
|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| urban  | —     | 0.255 | 0.220 | 0.022 | 0.040 | 0.145 | 0.016 | 0.229 | 0.228 | 0.168 | 0.173 | 0.002 |
| mill   | 0.255 | —     | 0.447 | 0.122 | 0.222 | 0.424 | 0.005 | 0.477 | 0.456 | 0.402 | 0.205 | 0.023 |
| for1   | 0.220 | 0.447 | —     | 0.077 | 0.415 | 0.364 | 0.145 | 0.669 | 0.571 | 0.467 | 0.259 | 0.056 |
| micro  | 0.022 | 0.122 | 0.077 | —     | 0.155 | 0.046 | 0.267 | 0.096 | 0.042 | 0.036 | 0.123 | 0.139 |
| for2   | 0.040 | 0.222 | 0.415 | 0.155 | —     | 0.229 | 0.354 | 0.395 | 0.347 | 0.250 | 0.172 | 0.163 |
| trees  | 0.145 | 0.424 | 0.364 | 0.046 | 0.229 | —     | 0.085 | 0.413 | 0.414 | 0.337 | 0.121 | 0.097 |
| phone  | 0.016 | 0.005 | 0.145 | 0.267 | 0.354 | 0.085 | —     | 0.081 | 0.104 | 0.097 | 0.185 | 0.208 |
| coins  | 0.229 | 0.477 | 0.669 | 0.096 | 0.395 | 0.413 | 0.081 | —     | 0.623 | 0.469 | 0.284 | 0.174 |
| cars   | 0.228 | 0.456 | 0.571 | 0.042 | 0.347 | 0.414 | 0.104 | 0.623 | —     | 0.471 | 0.296 | 0.151 |
| video  | 0.168 | 0.402 | 0.467 | 0.036 | 0.250 | 0.337 | 0.097 | 0.469 | 0.471 | —     | 0.214 | 0.154 |
| daisy  | 0.173 | 0.205 | 0.259 | 0.123 | 0.172 | 0.121 | 0.185 | 0.284 | 0.296 | 0.214 | —     | 0.169 |
| park   | 0.002 | 0.023 | 0.056 | 0.139 | 0.163 | 0.097 | 0.208 | 0.174 | 0.151 | 0.154 | 0.169 | —     |